

# Predictors of Opioid Misuse Disorder in the US

Final Project

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## 1 Abstract

 Note

**Naomi to complete.** Use 250 words or less.

Background: Opioid addiction is a public health crisis that has affected countless lives.

Research Question: What factors are important for predicting prevalence rate of opioid use disorder (OUD)?

Methods: We downloaded data on the prevalence of pain reliever misuse in each state from the National Survey on Drug Use and Health from the Substance and Mental Health Services Administration for all available years (2016–2019 and 2022)...[TBD]

Results: [TBD]

Conclusion: [TBD]

## 2 Key words

Opioid Use Disorder, Regression Analysis, Public Health, Substance Use Disorder

## 3 Introduction

Opioid addiction is a public health crisis that has affected countless lives. The United States is experiencing a public health crisis of almost unprecedented scale: an epidemic of opioid use disorder (OUD) and related overdose deaths, the magnitude of which has increased exponentially in the past decades (Manchester & Leshner, 2019). In just one year, the age-adjusted rate of drug overdose deaths increased 31% (from 21.6 in 2019 per 100k population to 28.3 in 2020) (Hedegaard et al., 2021). In

this project, we investigate what factors are important for predicting the prevalence rate of OUD across US states.

## 4 Literature review

### Note

**Alex to complete.** Discuss how other researchers have addressed similar problems, what their achievements are, and what the advantage and drawbacks of each reviewed approach are. Explain how your investigation is similar or different to the state-of-the-art. Please cite the relevant papers where appropriate.

Example citation (Volkow et al., 2019).

## 5 Methodology

### Note

Discuss the key aspects of your problem, data set and regression model(s). Given that you are working on real-world data, explain at a high-level your exploratory data analysis, how you prepared the data for regression modeling, your process for building regression models, and your model selection.

We use data on the state-level prevalence of pain reliever misuse and demographic characteristics from the National Survey on Drug Use and Health (NSDUH) from the Substance and Mental Health Services Administration for 2018-2024 (Substance Abuse and Mental Health Services Administration (SAMHSA)'s Data Analysis System (DAS), 2018-2024).

### 5.1 Exploratory Data Analysis (Naomi)

We performed exploratory data analysis to understand the distribution of our outcome variable (prevalence of pain reliever misuse) and the relationships between our predictors and the outcome. We visualized the distribution of the outcome variable across states and years, and we examined correlations between predictors and the outcome variable.

## 5.2 Data Preparation methods (Alex)

## 5.3 Assumptions (Nana)

## 5.4 Build models methods (Richie)

## 5.5 Select models methods (Said)

# 6 Experimentation and Results

### Note

Describe the specifics of what you did (data exploration, data preparation, model building, model selection, model evaluation, etc.), and what you found out (statistical analyses, interpretation and discussion of the results, etc.).

```
# Read in all CSVs from the data directory and store them in a list
filepaths <- list.files(path = "data", pattern = "*.csv", full.names = TRUE)

# Helper to extract the "YY-YY" year substring from filenames
extract_year_tag <- function(path) {
  fname <- basename(path)
  tag <- stringr::str_extract(fname, "\\d{2}-\\d{2}")
  if (is.na(tag)) tag <- gsub("\\.csv$", "", fname)
  tag
}

# Read each file and add a 'year' column based on the filename (e.g. "18-19")
data_list <- lapply(filepaths, function(fp) {
  df <- readr::read_csv(fp, show_col_types = FALSE) |>
    janitor::clean_names()

  # Get name of first column (variable of interest)
  first_col <- names(df)[1]

  # Transform each df.
  df <- df |>
    # Filter out redundant ("Overall") rows
    filter(
      state_us_abbreviation != "Overall") |>

  # Calculate precise percentages
```

```

group_by(state_us_abbreviation) |>
mutate(
  denominator = weighted_count[column_percent == 1]
) |>
ungroup() |>
mutate(
  percent = weighted_count / denominator
) |>

# Get subset of columns
select(1, state_us_abbreviation, percent) |>

# Pivot first column wider
mutate(
  !!first_col := paste0(first_col, "_", as.character(.data[[first_col]])),
) |>
pivot_wider(names_from = 1, values_from = percent) |>

# Drop columns with all the same values
select(where(~ n_distinct(.) > 1)) |>

# Add year column
mutate(
  year = extract_year_tag(fp)
) |>

# Clean names
janitor::clean_names()

df
})

# Get list of data years
years <- sapply(filepaths, extract_year_tag) |> unique()

# Iterate across years, join data frames in data_list by state and year within each year.
yearly_joined_data <- lapply(years, function(yr) {
  year_dfs <- data_list |>
  purrr::keep(~ dplyr::n_distinct(.x$year) == 1 && .x$year[[1]] == yr)

```

```

if (length(year_dfs) == 0) return(NULL)
if (length(year_dfs) == 1) return(year_dfs[[1]])

Reduce(function(x, y) full_join(x, y, by = c("state_us_abbreviation", "year")), year_dfs)
})

# Append yearly_joined_data across all years into a single data frame
# Filter out NULL values first to avoid bind_rows errors
df <- yearly_joined_data |>
  purrr::discard(is.null) |>
  bind_rows()

df_renamed <- df |>
  rename(
    state = state_us_abbreviation,

    age_12_17 = rc_adult_age_category_recode_5_12_17_years_old,
    age_18_25 = rc_adult_age_category_recode_1_18_25_years_old,
    age_26_34 = rc_adult_age_category_recode_2_26_34_years_old,
    age_35_49 = rc_adult_age_category_recode_3_35_49_years_old,
    age_50_plus = rc_adult_age_category_recode_4_50_or_older,

    medicaid_coverage = covered_by_medicaid_chip_1_yes,

    less_than_high_school = rc_education_categories_1_less_high_school,
    high_school = rc_education_categories_2_high_school_grad,
    some_college = rc_education_categories_3_some_coll_assoc_dg,
    college_grad = rc_education_categories_4_college_graduate,

    emp_full_time = employment_status_imputation_revised_1_employed_full_time,
    emp_part_time = employment_status_imputation_revised_2_employed_part_time,
    emp_unemployed = employment_status_imputation_revised_3_unemployed,

    pain_relievers_misuse = rc_pain_relievers_past_year_misuse_1_misused_within_the_past_year,

    race_nh_white = rc_race_recode_4_levels_1_white_not_hispanic,
    race_nh_black = rc_race_recode_4_levels_2_black_not_hispanic,
    race_hisp = rc_race_recode_4_levels_4_hispanic,
    race_other = rc_race_recode_4_levels_3_other_or_multiple_not_hispanic,

```

```

poverty = rc_poverty_level_new_inc_percent_of_us_census_poverty_threshold_1_living_in_pover
poverty_2x = rc_poverty_level_new_inc_percent_of_us_census_poverty_threshold_2_income_up_to
poverty_2x_plus = rc_poverty_level_new_inc_percent_of_us_census_poverty_threshold_3_income
) |>

select(
  state,
  year,
  pain_relievers_misuse,
  age_12_17, age_18_25, age_26_34, age_35_49, age_50_plus,
  medicaid_coverage,
  less_than_high_school, high_school, some_college, college_grad,
  emp_full_time, emp_part_time, emp_unemployed,
  race_nh_white, race_nh_black, race_hisp, race_other,
  poverty, poverty_2x, poverty_2x_plus
)

```

## 6.1 Exploratory Data Analysis (Naomi)

The data set contains 204 state-year rows and 23 features, and is 100 percent complete. The data are all numeric, and the target variable is `pain_relievers_misuse`, which represents the percentage of individuals who misused pain relievers in the past year. The other variables represent various demographic and socioeconomic characteristics of people surveyed in the state and year. See Table 1 for descriptive characteristics of the data.

```

tbl_summ_stats <- df_renamed |> skim_without_charts() |> janitor::clean_names() |> as_tibble()
  rename(Variable = skim_variable, "Completeness rate" = complete_rate, "Mean" = numeric_mean)
  select(Variable, "Completeness rate", Mean, Min, P25, P50, P75, Max) |>
  mutate(across(where(is.numeric), ~ signif(., 2))) |> knitr::kable()

tbl_summ_stats

```

Table 1: Summary statistics for all variables in the dataset

| Variable              | Completeness rate | Mean  | Min    | P25   | P50   | P75   | Max   |
|-----------------------|-------------------|-------|--------|-------|-------|-------|-------|
| state                 | 1                 | NA    | NA     | NA    | NA    | NA    | NA    |
| year                  | 1                 | NA    | NA     | NA    | NA    | NA    | NA    |
| pain_relievers_misuse | 1                 | 0.033 | 0.0120 | 0.026 | 0.032 | 0.039 | 0.058 |
| age_12_17             | 1                 | 0.091 | 0.0540 | 0.085 | 0.091 | 0.096 | 0.120 |
| age_18_25             | 1                 | 0.120 | 0.0980 | 0.120 | 0.120 | 0.130 | 0.160 |

Table 1: Summary statistics for all variables in the dataset

| Variable              | Completeness rate | Mean  | Min    | P25   | P50   | P75   | Max   |
|-----------------------|-------------------|-------|--------|-------|-------|-------|-------|
| age_26_34             | 1                 | 0.140 | 0.1200 | 0.130 | 0.140 | 0.140 | 0.250 |
| age_35_49             | 1                 | 0.220 | 0.2000 | 0.210 | 0.220 | 0.230 | 0.260 |
| age_50_plus           | 1                 | 0.430 | 0.3100 | 0.410 | 0.430 | 0.440 | 0.500 |
| medicaid_coverage     | 1                 | 0.190 | 0.0500 | 0.150 | 0.190 | 0.230 | 0.370 |
| less_than_high_school | 1                 | 0.084 | 0.0460 | 0.068 | 0.079 | 0.097 | 0.160 |
| high_school           | 1                 | 0.250 | 0.1300 | 0.220 | 0.250 | 0.270 | 0.370 |
| some_college          | 1                 | 0.280 | 0.1500 | 0.270 | 0.280 | 0.300 | 0.380 |
| college_grad          | 1                 | 0.290 | 0.1500 | 0.260 | 0.290 | 0.330 | 0.590 |
| emp_full_time         | 1                 | 0.440 | 0.2700 | 0.420 | 0.440 | 0.460 | 0.570 |
| emp_part_time         | 1                 | 0.130 | 0.0790 | 0.120 | 0.130 | 0.140 | 0.190 |
| emp_unemployed        | 1                 | 0.042 | 0.0170 | 0.033 | 0.041 | 0.050 | 0.075 |
| race_nh_white         | 1                 | 0.680 | 0.2100 | 0.580 | 0.710 | 0.800 | 0.940 |
| race_nh_black         | 1                 | 0.110 | 0.0042 | 0.032 | 0.072 | 0.150 | 0.440 |
| race_hisp             | 1                 | 0.120 | 0.0150 | 0.051 | 0.092 | 0.140 | 0.480 |
| race_other            | 1                 | 0.089 | 0.0230 | 0.047 | 0.065 | 0.095 | 0.680 |
| poverty               | 1                 | 0.150 | 0.0710 | 0.110 | 0.140 | 0.170 | 0.270 |
| poverty_2x            | 1                 | 0.190 | 0.0900 | 0.170 | 0.190 | 0.220 | 0.270 |
| poverty_2x_plus       | 1                 | 0.660 | 0.5000 | 0.620 | 0.670 | 0.720 | 0.780 |

```

# Get correlation matrix
df_renamed_cor <- df_renamed |>
  select(-state, -year) |>
  cor(use = "pairwise.complete.obs")

# Suppress small correlations by setting them to NA so corrplot leaves them blank
df_renamed_cor[abs(df_renamed_cor) < 0.05] <- NA

corrplot(df_renamed_cor, method = "circle", type = "upper", tl.cex = 0.4, insig = "blank", na.lty = "n")

most_corr_w_target <- df_renamed_cor |>
  as.data.frame() |>
  rownames_to_column(var = "Variable") |>
  filter(Variable != "pain_relievers_misuse") |>
  transmute(
    Variable,
    Correlation = pain_relievers_misuse,
  )

```

```

Abs_Correlation = abs(pain_relievers_misuse)
) |>
arrange(desc(Abs_Correlation)) |>
filter(Abs_Correlation > 0.05) |>
pull(Variable)

```

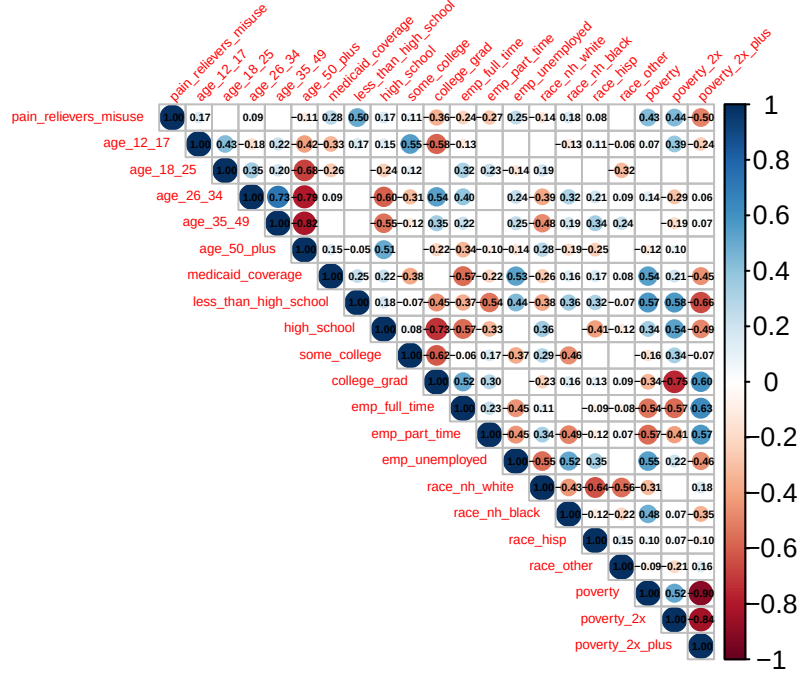


Figure 1: Correlation matrix of all non-ID variables in the dataset. Correlations  $< 0.05$  are suppressed.

The correlation between all variables are shown in Figure 1. The following predictors are most correlated with the target variable: less\_than\_high\_school, poverty\_2x\_plus, poverty\_2x, poverty, college\_grad, medicaid\_coverage, emp\_part\_time, emp\_unemployed, emp\_full\_time, race\_nh\_black, high\_school, age\_12\_17, race\_nh\_white, age\_50\_plus, some\_college, age\_26\_34, race\_hisp, which may be predictors of pain reliever misuse.

## 6.2 Results of Data Preparation (Alex)

## 6.3 Results of Assumption checks (Nana)

## 6.4 Results of Building models (Richie)

## 6.5 Results of model selection (Said)

# 7 Discussion and Conclusions (Naomi)

*Conclude your findings, limitations, and suggest areas for future work.*



## References

- Hedegaard, H., Miniño, A. M., Spencer, M. R., & Warner, M. (2021). *Drug overdose deaths in the united states, 1999–2020* (No. 428). <http://dx.doi.org/10.15620/cdc:112340>
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- Substance Abuse and Mental Health Services Administration (SAMHSA)’s Data Analysis System (DAS). (2018-2024). *NSDUH: 2-year restricted-use data (2018-2019, 2021-2022, 2022-2023, 2023-2024)*. [https://datatools.samhsa.gov/das/nsduh/2022/nsduh-2022-2023-rd02yr/crosstab?row=RACE4&column=STUSAB&weight=DASWT\\_1](https://datatools.samhsa.gov/das/nsduh/2022/nsduh-2022-2023-rd02yr/crosstab?row=RACE4&column=STUSAB&weight=DASWT_1)
- Volkow, N. D., Jones, E. B., Einstein, E. B., & Wargo, E. M. (2019). Prevention and treatment of opioid misuse and addiction: A review. *JAMA Psychiatry*, 76(2), 208–216. <https://doi.org/10.1001/jamapsychiatry.2018.3126>